

12.421/12.621 Lecture 14: Synthetic Aperture Radar, Part 1

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Topics covered

- History and overview of radar
- Basics of imaging radar
- Fundamentals of synthetic aperture radar

1 Overview of radar

- Originally an acronym “RADio Detection And Ranging” (now a word)
- Perhaps the first system we might recognize as a radar was developed by Hertz ~1880s to test Maxwell’s EM theory – demonstrated that reflections from various objects were detectable
- Military needs in the early 20th century drove significant development
 - uses extend from detection of ships and planes to weapons development (e.g., proximity fuses)
- Some of the first images of the ground taken from an airborne radar were produced by MIT Radiation Laboratory, precursor to Lincoln Labs
- These early systems used a method known as side-looking airborne radar (SLAR), which “scans” the surface using a “real-aperture radar”
 - Main disadvantage to this approach is the poor azimuth resolution (as we shall see)
 - Az. resolution inversely proportional length of antenna (and proportional to distance to target), “good” resolution (<100s of meters) required huge antenna, too large to fly on aircraft or spacecraft
- In 1952, Carl Wiley, working for Goodyear Aircraft Corporation, invented a method called “Doppler beam-sharpening” that synthesized a large aperture using a much smaller aperture.
 - Dramatically improved azimuth resolution → proportional to antenna length, independent of range
 - Technique now known as synthetic aperture radar (SAR)
- Operational SAR systems started to appear in the 1960s and continue to be extremely important to Earth science and defense and other industrial applications
- Aside from high-resolution, SAR is valuable because it operates day and night and in virtually any weather – sees through clouds, smoke, pollution, etc.

2 Imaging radar fundamentals – how does it work?

- Radar is an active sensor – it measures range to a target by transmitting an EM signal and receiving the echo reflected by the object
- Radar instruments measure time of flight, thus range to target is simply

$$R = \frac{1}{2}u_p t \quad \rightarrow \text{hereafter we assume air} \Rightarrow u_p \approx c \quad (1)$$

where the $\frac{1}{2}$ accounts for two-way travel

- Unlike optical images that are acquired by recording a lens-projected 2D image and an array of detectors, a single moving detector, or chemical processes (film), Radars form images by scanning surfaces, using the platform velocity to acquire long strip images.

★ Strips often called ‘swaths’ or ‘tracks’

- Due to strip-mapping, we distinguish directions in radar imagery according to the platform velocity vector:
 - Range (a.k.a. cross-track) is perpendicular to velocity
 - Azimuth (a.k.a. along-track) is parallel to velocity
- In range direction, radar echoes are separated using the time between the echoes from different scatterers

★ True for both real and synthetic apertures

- In azimuth direction, surface pixels are separated according to the angular size of the radar footprint (in real apertures) or the Doppler history (synthetic aperture)

2.1 Imaging geometry and the radar footprint

- Pulses transmitted from radar as platform moves (Figure 1)
- Each pulse illuminates an ellipse on the ground

– defined in range:

$$\theta_r \approx \frac{\lambda}{W} \quad \text{and} \quad \underbrace{S_r \approx \frac{h\theta_r}{\cos^2 \theta}}_{\text{flat-Earth assumption}} = \frac{\lambda h}{W \cos^2 \theta} \quad (2)$$

– in azimuth:

$$\theta_a \approx \frac{\lambda}{L} \quad \text{and} \quad S_a \approx R\theta_a \approx \frac{\lambda h}{L \cos \theta} \quad (3)$$

E.g. for a spacecraft: $h = 800$ km, $\lambda = 23$ cm (L-band), $L = 12$ m, $\theta = 20^\circ \rightarrow S_a = 16$ km

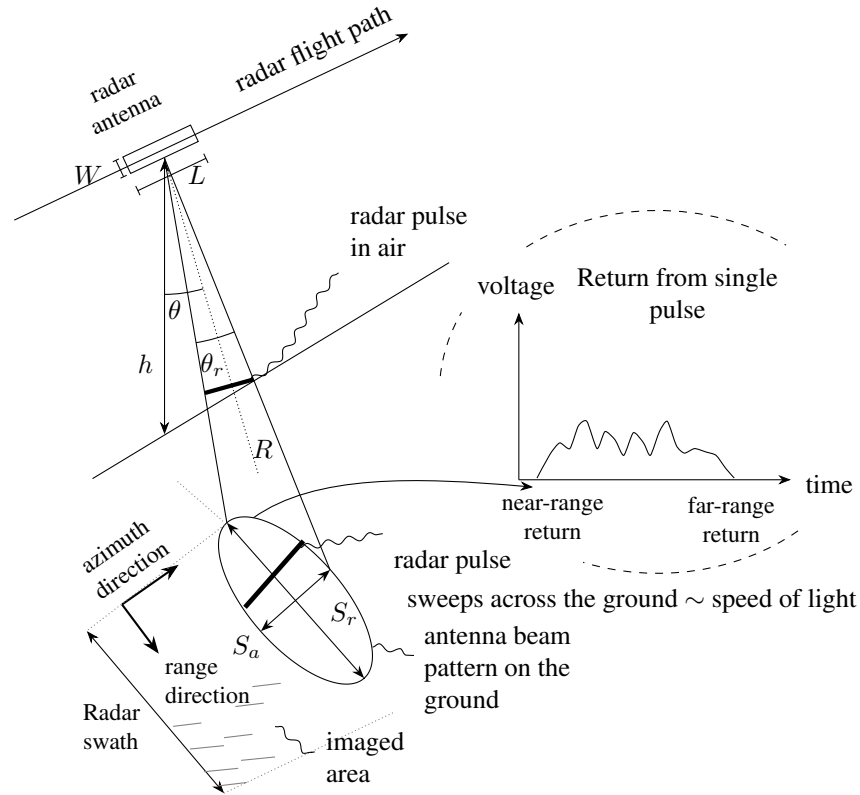


Figure 1: Imaging geometry of a side-looking radar: the antenna (width W , length L) at altitude h illuminates an elliptical footprint $S_r \times S_a$ on the ground at range R and look angle θ ; each pulse sweeps across the footprint, and the echo from a single pulse (inset) runs from the near-range to the far-range return (cf. van Zyl (2011), Fig. 1-2).

2.2 Azimuth resolution of a real-aperture radar

- For a real-aperture radar, $S_a = X_a \equiv$ azimuth resolution
 - Resolution defines smallest separation between two objects that can be uniquely detected
 - Note example on previous page $\rightarrow X_a = 16$ km is horrible resolution
 - Even if we decrease λ to 2 cm (Ku-band) and made $h = 200$ km (unreasonably low altitude for a satellite), $X_a = 360$ m, which is still pretty lousy resolution for most science applications
 - Poor resolution limits application of real-aperture radar for RS applications, though there are some notable exceptions $\rightarrow$ terrestrial radar
 - Synthetic aperture techniques will dramatically improve azimuth resolution, and will lead to azimuth resolution equal to half the length of the antenna!!! Will discuss how in a bit

2.3 Range resolution

- What is range resolution?
 - First, distinguish slant range and ground range (Figure 2)

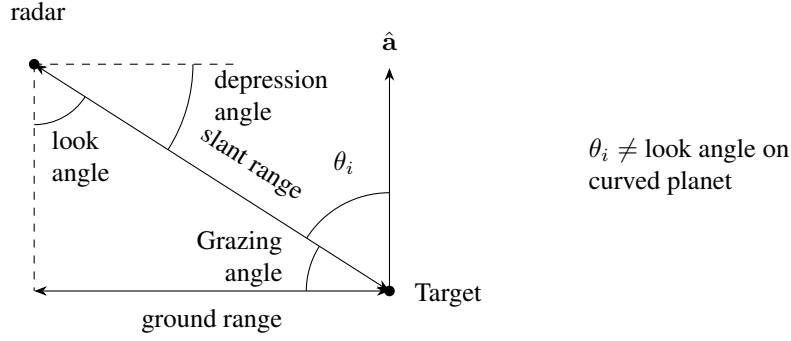


Figure 2: Slant range, ground range and the angles that describe the imaging geometry (cf. van Zyl (2011), Fig. 1-3). Neglecting the curvature of the Earth, and for a surface without topography, the look angle equals the incidence angle θ_i .

- slant range $\equiv$ distance from radar to scatterer (target)
- ground range $\equiv$ distance from nadir point to scatterer along a smooth (not necessarily flat) surface
- Now consider two point targets separated by a slant range distance $X_r \Rightarrow$ echoes will be separated in time

$$\Delta t = \frac{2X_r}{c} \quad (4)$$

2.3.1 Pulse length and resolution

- Radars transmit pulses, i.e., short bursts of energy
- For a pulse w/ constant frequency, two scatterers can be distinguished if the leading edge of the pulse from 2nd scatterer is received after the trailing edge of pulse from 1st scatterer (Figure 3)

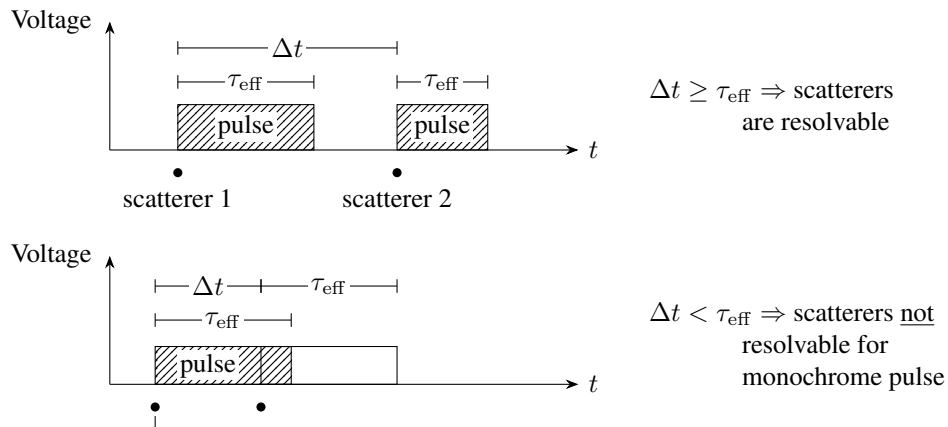


Figure 3: Echoes from two point scatterers for a constant-frequency (monochromatic) pulse of length τ_{eff} : resolvable when their separation in time is $\Delta t \geq \tau_{\text{eff}}$ (top), not resolvable when $\Delta t < \tau_{\text{eff}}$ (bottom) (cf. van Zyl (2011), Fig. 1-5).

- Thus, smallest separable $\Delta t = \tau_{\text{eff}}$, where $\tau_{\text{eff}} \equiv$ pulse length

$$\Rightarrow \frac{2X_r}{c} = \Delta t = \tau_{\text{eff}} \Rightarrow X_r = \frac{c\tau_{\text{eff}}}{2} \quad (5)$$

- What does this mean for ground range resolution?
- Two point targets w/ ground range separation X_g

$$\Delta t = \frac{2X_g \sin \theta_i}{c} \quad (6)$$

- Based on argument above,

$$X_g = \frac{c\tau_{\text{eff}}}{2 \sin \theta_i} \longrightarrow \boxed{\text{No dependence on distance to target!!!}} \quad (7)$$

- Note the dependence of ground range resolution on incidence angle $\Rightarrow$ for slopes facing radar, θ_i decreases $\Rightarrow X_g$ increases (and vice versa)

★ ground range resolution varies dramatically in mountains

2.3.2 Noise and the pulse length

- It would appear at this point that improving resolution is a simple matter of reducing the pulse duration τ_{eff}
- Why would we not want to do that?
- A. Noise
 - Recall lecture on radiation: Planck's law tells us that any material w/ $T > 0$ K emits radiation at a range of frequencies
 - Radars must operate at $T > 0$ K $\Rightarrow$ there will always be some thermal noise power

$$P_N = k_B T \Delta f \quad (8)$$

$k_B \equiv$ Boltzmann's const

$\Delta f \equiv$ bandwidth of instrument (not necessarily same as pulse bandwidth)

- Signal to noise ratio is then simply received power P_r divided by P_N

$$\text{SNR} = \frac{P_r}{P_N} \quad (9)$$

- To increase SNR, want to maximize energy in each pulse $\rightarrow$ can be done by increasing peak power or by using a longer pulse
- Peak power usually limited by power supply (esp. on satellites) $\Rightarrow$ increasing SNR depends on lengthening pulse duration
- There is a direct conflict between SNR and range resolution!
 - longer pulse = higher SNR (good), lower resolution (bad)
 - shorter pulse = lower SNR (bad), higher resolution (good)

2.4 Frequency-modulated pulses (chirp)

- The solution to SNR–resolution conundrum is to abandon monochromatic pulses and to modulate the frequencies instead
- ‘chirp’ – signal whose frequency changes in time
- Radar generally uses a linear chirp

$$\Rightarrow f(t) = f_0 + \frac{B}{\tau_p} t \quad \text{for } -\frac{\tau_p}{2} \leq t \leq \frac{\tau_p}{2} \quad (10)$$

where

$$B = |(f + \Delta f) - f| = |\Delta f| \equiv \text{pulse bandwidth}$$

$$\tau_p \equiv \text{true pulse duration (different from effective pulse duration } \tau_{\text{eff}} = 1/B)$$

- Note that $B = 1/\tau_{\text{eff}}$ is a property of radar instrument while τ_p is the true pulse duration
 - ★ Range resolution $X_r \sim X_g \sim \frac{1}{B}$
 - ★ SNR scales w/ τ_p
 - ★ Using frequency modulated (FM) pulses, possible to construct system and pulse w/ high bandwidth ($\Rightarrow$ fine resolution) and long pulse duration τ_p ($\Rightarrow$ high SNR)
- Signal amplitude of chirped signal

$$A(t) \sim \cos\left(\omega_0 t + \frac{\pi B}{\tau_p} t^2\right) \quad \text{for } -\frac{\tau_p}{2} \leq t \leq \frac{\tau_p}{2} \quad (11)$$

- Matched filtering allows for recovery of fine-resolution features from pulse train $\rightarrow$ range compression

2.5 Range compression

- Echoes from different scatterers will overlap in time, but $\therefore$ FM pulse echoes will have different frequencies $\Rightarrow$ can be separated by appropriate filtering
- Matched filters often used in radar processing
 - Idea is simple: the signal received from a scatterer will be a scaled replica of the transmitted signal delayed by time $t = 2R/c$
 - Therefore, simply convolve return signal w/ transmit signal
 - To illustrate, suppose a transmit signal

$$A(t) = A_0 e^{j(\omega_0 t + \pi B t^2 / \tau_p + \phi)} \quad \text{for } -\frac{\tau_p}{2} \leq t \leq \frac{\tau_p}{2} \quad (12)$$

(the physical signal is the real part of $A(t)$, i.e., the chirp of Eq. (11)) and the received signal is

$$V_r(t) = E_r A\left(t - \frac{2R}{c}\right) \quad (13)$$

then the compressed radar signal is the convolution

$$\begin{aligned}
 V_o(t) &= \int_{-\infty}^{\infty} A^*(\xi - t) V_r(\xi) d\xi \\
 &\approx \tau_p |A_0|^2 E_r e^{j(\omega_0 t - 2kR)} \underbrace{\frac{\sin(\pi B(t - 2R/c))}{\pi B(t - 2R/c)}}_{\text{norm. sinc function}}
 \end{aligned} \tag{14}$$

- Peak of compressed pulse occurs at $t = \frac{2R}{c}$ and it has a half power width of approximately $\frac{1}{B}$, consistent w/ our previous observation that range resolution $\sim \frac{1}{B}$

3 Synthetic aperture radar (SAR)

- ‘Synthetic aperture radar’ utilizes imaging radar data collected from a moving platform and a special processing technique that stitches together many pulses to synthesize a much larger antenna (aperture)
- Key is to recognize that a target will be seen by the radar at different points in the radar footprint during different pulses $\Rightarrow$ slight differences in angles w.r.t. velocity vector $\Rightarrow$ Doppler shift
- Using Doppler effect, multiple targets can be separated in the azimuth direction $\Rightarrow$ greatly improving azimuth resolution for a given radar instrument
- Use of Doppler effect to compress signals in azimuth is the difference between SAR and real-aperture radar
 - $\hookrightarrow$ range resolution and compression discussed before are the same for SAR and real-aperture radars
- So, SAR relies on Doppler effect and sensor motion (Figure 4)

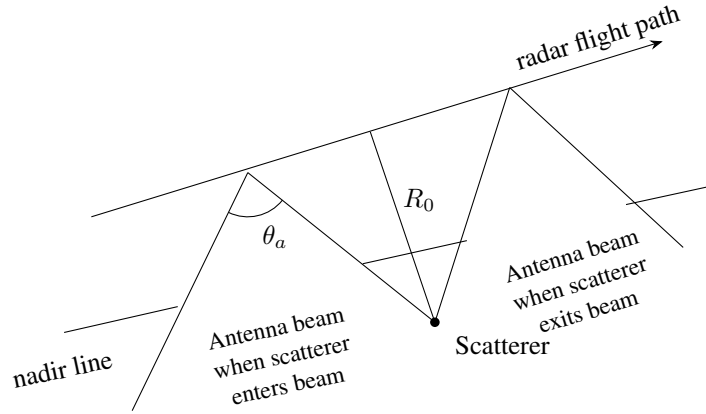


Figure 4: The synthetic aperture: a scatterer at range of closest approach R_0 is observed for as long as it remains in the antenna beam of azimuth beamwidth θ_a , from the pulse at which it enters the beam to the pulse at which it exits (cf. van Zyl (2011), Fig. 1-7).

3.1 Doppler history of a point scatterer

- Distance between antenna and scatterer as a function of time s (a.k.a. slow time)

$$R(s) = \sqrt{R_0^2 + v^2 s^2} \quad (15)$$

$R_0 \equiv$ minimum distance from flight path to scatterer

$v \equiv$ platform velocity

- Usually in RS, $vs \ll R_0$

$$\Rightarrow R(s) \approx R_0 + \frac{v^2}{2R_0} s^2 \quad (16)$$

- The phase after range compression (see expression on previous pages) is given as

$$\phi(s) = -\frac{4\pi R(s)}{\lambda} \approx -\frac{4\pi R_0}{\lambda} - \frac{2\pi v^2}{R_0 \lambda} s^2 \quad (17)$$

- The instantaneous frequency is simply

$$f(s) = \frac{1}{2\pi} \frac{\partial \phi(s)}{\partial s} = -\frac{2v^2}{R_0 \lambda} s \quad (s = 0 \text{ when } R = R_0) \quad (18)$$

- Note that this, Eq. (18), is expression of linear chirp, just like we saw for FM pulse
- To find Doppler bandwidth $\rightarrow$ maximum time we can use for signal integration $\rightarrow$ amount of time scatterer is in antenna footprint

3.2 SAR azimuth resolution

- As discussed before, for an antenna w/ along-track length L ,

$$\theta_a = \frac{\lambda}{L} \Rightarrow s_{\text{tot}} = \frac{\lambda R_0}{Lv} \quad (19)$$

- Doppler bandwidth

$$B_D = |f(s_{\text{tot}})| = \frac{2v}{L} \quad (20)$$

- Using matched filter approach (as with range compression), the compressed signal will have a duration $1/B_D$

$$\Rightarrow \text{Azimuth resolution} \quad \boxed{X_a^{\text{SAR}} = \frac{v}{B_D} = \frac{L}{2}} \quad (21)$$

$\rightarrow$ Amazing! only a function of length of physical antenna!!!

- SAR azimuth resolution

$$X_a^{\text{SAR}} = \frac{L}{2} \rightarrow \text{compare w/ } X_a^{\text{RAR}} = \frac{R\lambda}{L} \quad (22)$$

- Smaller antenna = finer resolution? Yes because smaller antenna result in broader footprints $\Rightarrow$ longer s_{tot} (time in beam) $\Rightarrow$ higher Doppler bandwidth

- No range dependence on azimuth resolution because longer range gives broader footprint $\Rightarrow$ longer s_{tot} (longer synthetic aperture) that cancels range effect on resolution
- Since radar is pulsed, it's necessary to ensure pulse repetition frequency (PRF) sufficient to sample Doppler

$$\Rightarrow \text{PRF} \geq B_D = \frac{2v}{L} \quad (23)$$

$\Rightarrow$ one sample should be taken when the antenna has moved at most $\frac{L}{2}$

- Next time, image artifacts, ambiguities and noise, then intro to InSAR

Variables

Units are SI; “arb.” marks signals in arbitrary units. Values are given for universal constants and for the examples in the notes.

Symbol	Description	Units	Value
<i>Constants</i>			
c	speed of light in a vacuum	m/s	2.998×10^8
k_B	Boltzmann constant	J/K	1.381×10^{-23}
j	imaginary unit	–	$\sqrt{-1}$
<i>Imaging geometry</i>			
R	range (slant range) from the antenna to a target	m	$\frac{1}{2}u_p t$
t	time (round-trip travel time; time within a pulse)	s	–
u_p	propagation (phase) velocity	m/s	$\approx c$ in air
λ	radar wavelength	m	23 cm (L-band), 2 cm (Ku-band) in the examples
W, L	width and (along-track) length of the antenna	m	$L = 12$ m in the example
h	altitude of the radar above the surface	m	800 km, 200 km in the examples
θ	look angle (from the vertical to the beam center)	rad	20° in the example
θ_r, θ_a	beamwidths in range and azimuth	rad	$\lambda/W, \lambda/L$
S_r, S_a	range and azimuth extents of the footprint on the ground	m	$\lambda h/(W \cos^2 \theta),$ $\lambda h/(L \cos \theta)$
θ_i	incidence angle (from the local normal)	rad	$= \theta$ (flat Earth, no topography)
$\hat{\mathbf{a}}$	unit normal to the surface	dimensionless	–
<i>Resolution</i>			
X_a	azimuth resolution (real aperture: S_a)	m	16 km, 360 m in the examples
$X_a^{\text{RAR}}, X_a^{\text{SAR}}$	azimuth resolution of a real- and a synthetic-aperture radar	m	$R\lambda/L; L/2$
X_r, X_g	slant-range and ground-range resolutions (separations)	m	$c\tau_{\text{eff}}/2; c\tau_{\text{eff}}/(2 \sin \theta_i)$
Δt	separation in time of two echoes	s	–
τ_{eff}	effective pulse length (duration)	s	$1/B$
τ_p	true pulse duration	s	–
<i>Noise</i>			
T	(total equivalent noise) temperature	K	–
Δf	bandwidth of the instrument (L8: frequency excursion of the chirp, $ \Delta f = B$)	Hz	–
P_N	thermal noise power	W	$k_B T \Delta f$
P_r	received power	W	–
SNR	signal-to-noise ratio	dimensionless	P_r/P_N

Symbol	Description	Units	Value
<i>Chirp and range compression</i>			
$f, f(t)$	frequency; instantaneous frequency of the chirp	Hz	$f_0 + Bt/\tau_p$
f_0, ω_0	center (carrier) frequency and angular frequency	Hz; rad/s	$\omega_0 = 2\pi f_0$
B	pulse (chirp) bandwidth	Hz	$1/\tau_{\text{eff}}$
$A(t)$	transmitted (chirp) signal	arb.	–
A_0, ϕ	amplitude and phase constant of $A(t)$	arb.; rad	–
$V_r(t)$	received signal	arb.	$E_r A(t - 2R/c)$
$V_o(t)$	compressed (matched-filter output) signal	arb.	–
E_r	field received from a point scatterer (scale factor of the replica)	arb.	–
ξ	integration variable (time)	s	–
k	wavenumber, $2\pi/\lambda$	rad/m	–
<i>Synthetic aperture</i>			
s	slow time (along the flight path)	s	0 at closest approach
$R(s), R_0$	range to the scatterer; range at closest approach	m	$\sqrt{R_0^2 + v^2 s^2}$
v	platform velocity	m/s	–
$\phi(s)$	phase after range compression	rad	$-4\pi R(s)/\lambda$
$f(s)$	instantaneous (Doppler) frequency	Hz	$-2v^2 s/(R_0 \lambda)$
s_{tot}	time the scatterer is in the beam	s	$\lambda R_0/(Lv)$
B_D	Doppler bandwidth	Hz	$2v/L$
PRF	pulse repetition frequency	Hz	$\geq 2v/L$